Field Hours

CSCE 727 Algorithmic Foundations of Big Data Computing

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

One pass and a majority counter

You process a stream when the data will not fit in working memory. Items arrive once, in order, and you may store only a small summary. A counter of the length uses a logarithmic number of bits and, after 1000 items, holds 1000. The interesting summaries answer a harder question with fewer bits than the raw list. Before you design one, write the pass limit, here 1 pass, and the memory bound you are willing to spend.

You find a majority item with one counter and one candidate, the Boyer-Moore procedure. Start empty. On each item, if the counter is 0, adopt that item and set the counter to 1. If the item matches the candidate, add 1. Otherwise subtract 1. On the stream a, b, a, c, a, b, a of length 7, the candidate ends as a. A second look, or a promise, confirms that a occurs 4 times, which is more than 7/2.

One counter hunts a majority item.

1. At 0, adopt the item

2. A match adds 1

3. A mismatch subtracts 1

4. Check the survivor

A stream of 7 symbols

1. Stream a, b, a, c, a, b, a. Three mismatches return the counter to 0, and the last symbol sets the candidate to a with count 1.

2. In the raw stream, a occurs 4 times.

3. Half of 7 is 3.5, and 4 is larger, so a is a strict majority. The counter did not store b or c.

Problem 1. On this stream of length 7, how many times does a occur, and is that a strict majority?

Answer. a occurs 4 times. Half of 7 is 3.5, and 4 > 3.5, so yes.

Problem 2. After 1000 arrivals, what does a length counter hold, and does that alone name a majority item?

Answer. It holds 1000. No. The majority procedure also keeps a candidate, updated by adding 1 or subtracting 1.

Lecture 2 · 90 minutes

A count-min sketch

You estimate frequencies with a count-min sketch of 2 rows and 4 columns. Each row hashes items into its own 4 cells. An arrival adds 1 to one cell in each row. The estimate is the minimum of the cells the item touches. That minimum cannot fall below the true count, because each of the item's arrivals was added, and other items only increase a cell. Width 4 is the memory per row.

You recover exact counts on a short stream when one row separates the items. The stream a, a, b, a has true counts 3 and 1. Row 1 hashes both items to column 0, so that cell holds 4. Row 2 hashes a to column 1, holding 3, and b to column 2, holding 1. The estimate for a is min(4, 3) = 3. The estimate for b is min(4, 1) = 1. The min operation discarded the collision.

Count-min keeps the smaller cell.

Hash into each row -> Add 1 on each row -> Take the minimum -> Stay at or above truth

Stream a, a, b, a

1. True counts are 3 for a and 1 for b. Row 1 collides and stores 4 in one cell.

2. Row 2 stores 3 under a and 1 under b.

3. The estimates are min(4, 3) = 3 for a and min(4, 1) = 1 for b. Both match the truth on this stream.

Problem 1. The two row cells for item a hold 6 and 4. What does count-min report, and can the true count be 5?

Answer. It reports min(6, 4) = 4. The true count cannot be 5, because every reported value is at least the true count.

Problem 2. In the worked stream, what did the colliding row store for a, and what estimate survived?

Answer. The colliding row stored 4. The other row stored 3, and the minimum kept 3.

Lecture 3 · 90 minutes

Trailing zeros and distinct items

You estimate distinct items without storing the set. Hash each item to bits and keep the largest number of trailing zeros seen so far. A hash ends with at least r trailing zeros with probability about 1 over 2 to the r, so an observed r suggests about 2 to the r distinct items. If your maximum is 3, the estimate is 8. One hash is a coarse sketch of that rare event.

You steady the estimate by keeping several independent hashes and combining them. Suppose 4 hashes report maximum trailing-zero counts 2, 3, 2, and 1. The corresponding raw estimates are 4, 8, 4, and 2. Their average is 18/4 = 4.5. The median of those four numbers is 4, which is less sensitive to the single 8. The mechanism is still the rare trailing-zero event. Averaging 4 sketches only reduces the noise around it.

Trailing zeros become a distinct-count estimate.

Keep the max trailing zeros

A max of 3 estimates 8

Raw estimates 4, 8, 4, 2

Average of the four is 4.5

Combine four raw estimates

1. Trailing-zero records 2, 3, 2, and 1 become raw estimates 4, 8, 4, and 2.

2. The average is (4 + 8 + 4 + 2) / 4 = 18/4 = 4.5.

3. Sorted, the two central values are 4 and 4, so the median is 4. The lone 8 moves the average only.

Problem 1. The maximum number of trailing zeros is 4. What distinct-count estimate does the power-of-two rule give?

Answer. 2 to the power 4 equals 16.

Problem 2. Raw estimates are 4, 8, 4, and 2. What is their average?

Answer. (4 + 8 + 4 + 2) / 4 = 18/4 = 4.5.

Lecture 4 · 90 minutes

A reservoir of size 1

You sample a stream of unknown length with a reservoir of size 1. Store the first item. When item t arrives, replace the stored item with probability 1/t. After 5 items, each item remains with probability 1/5. The first item survives step 2 with probability 1/2, step 3 with probability 2/3, step 4 with probability 3/4, and step 5 with probability 4/5. That product equals 1/5.

You can check the three-item case by enumerating the replacement coins. Items arrive as 10, then 20, then 30. You always store 10 at step 1. Item 20 enters with probability 1/2. Item 30 enters with probability 1/3, regardless of what is stored. The probability that 10 is still stored at the end is (1/2)*(2/3) = 1/3. The probability that 20 remains is (1/2)*(2/3) = 1/3. Item 30 is stored with probability 1/3.

Reservoir of size 1, one arrival at a time.

1. Store the first item

2. At step t, replace with 1/t

3. After 5 items each has chance 1/5

Items 10, 20, and 30

1. Item 20 replaces 10 with probability 1/2. Item 30 replaces the stored item with probability 1/3.

2. The probability that 10 survives both later steps is (1/2)*(2/3) = 1/3.

3. The probability that 20 is chosen and then kept is also (1/2)*(2/3) = 1/3, and 30 is stored with probability 1/3.

Problem 1. After 5 items, what is the probability that a reservoir of size 1 holds the third item?

Answer. 1/5. Each of the 5 items has the same survival probability.

Problem 2. Write the product that leaves the first of 5 items stored, and its value.

Answer. (1/2)*(2/3)*(3/4)*(4/5) = 1/5.

Lecture 5 · 90 minutes

A sublinear density test

You test a huge array with a sample whose size ignores the length. The bits are either all zeros or at least 1/4 ones. Draw 8 indices. If you see a one, reject the all-zero claim. When a quarter of the bits are ones, one draw misses with probability 3/4, so eight draws all miss with probability (3/4) to the 8. That equals 6561/65536, about 0.100. You catch the dense case about 9 times in 10.

You spend 8 probes rather than a full scan. One probe is a weak test: if only 1/4 of the bits are ones, it returns zero with probability 3/4. Moving from 1 probe to 8 drops the miss probability from 3/4 to about 0.100. The procedure can still accept a dense array, so it is Monte Carlo. A second independent batch of 8 probes is how you cut that error again.

Eight probes against an array that is one quarter ones.

One probe

Miss probability 3/4

Too weak as a test

Eight probes

Miss probability about 0.100

Detects about 9 in 10

Compute (3/4) to the 8

1. Each probe misses a 1/4-dense array with probability 3/4.

2. Eight independent misses have probability 3^8 / 4^8 = 6561/65536.

3. That is about 0.100, so the detection probability is about 0.900.

Problem 1. What is (3/4)^8 as an integer ratio, and about what decimal is it?

Answer. 6561/65536, about 0.100.

Problem 2. The array is all zeros and your 8 probes are all zeros. Do you reject the all-zero claim?

Answer. No. You reject only when a probe sees a one. Eight zeros are what an all-zero array always returns.

Lecture 6 · 90 minutes

One round of map and reduce

You aggregate by one round of map and reduce. A mapper emits key-value pairs from a local shard. A reducer receives every pair that shares its key and combines them. Take 6 tokens: cat, dog, cat, cat, dog, bird. Each token emits a pair such as (cat, 1). The cat reducer sums three 1s and emits 3. Dog emits 2 and bird emits 1. The shuffle moves exactly those 6 pairs.

You count rounds, because a second round is a new algorithm. Word count finished in 1 round: after the shuffle, each reducer already holds every pair for its key. Asking for the maximum among the three counts 3, 2, and 1 needs another round, and that round returns 3. The mechanism is the keyed shuffle. The resource you write down is the round count, 1 for the counts and 2 if you also want their maximum.

One shuffle produces the three counts.

Map tokens to pairs -> Shuffle by the key -> Sum the ones -> Emit 3, 2, and 1

Six tokens, three keys

1. The tokens cat, dog, cat, cat, dog, and bird emit 6 pairs.

2. Cat sums three 1s and emits 3. Dog emits 2. Bird emits 1.

3. That used 1 round. Taking the max of 3, 2, and 1 needs a second round and returns 3.

Problem 1. How many pairs move in the shuffle of these 6 tokens, and what count does cat get?

Answer. Six pairs move. Cat gets 3.

Problem 2. Counts 3, 2, and 1 already sit on three reducers. How many extra rounds produce their maximum, and what is it?

Answer. One extra round. The maximum of 3, 2, and 1 is 3.

Lecture 7 · 90 minutes

A second-moment sketch

You estimate the second frequency moment with random signs. Assign each distinct item a sign of plus 1 or minus 1, and keep the weighted sum Z of the arrivals. The square of Z is an unbiased estimate of F2, the sum of squared frequencies. Frequencies 2, 1, and 1 give F2 = 4 + 1 + 1 = 6. A single sign pattern can return a square other than 6. The expectation over the signs equals 6.

You average copies when one square is unlucky. Four sketches of the same frequencies return 4, 6, 10, and 4. Their average is (4 + 6 + 10 + 4) / 4 = 6, matching F2. The sketch that returned 10 was off by 4 on its own. You store each copy as one integer and a seed that rebuilds the signs. Four integers stand in for the frequency table of the whole stream.

Random signs, then an average of squares.

Frequencies 2, 1, and 1

Second moment equals 6

Squares 4, 6, 10, and 4

Their average equals 6

Average four sketches

1. The true second moment is 4 + 1 + 1 = 6.

2. Four independent squares return 4, 6, 10, and 4.

3. Their average is (4 + 6 + 10 + 4) / 4 = 24/4 = 6. The copy that returned 10 was off by 4 before the average.

Problem 1. Frequencies are 3 and 1. What is F2?

Answer. 3 squared plus 1 squared is 9 + 1 = 10.

Problem 2. Four sketches return 4, 6, 10, and 4 on a stream whose F2 is 6. What is the average, and how far off was the 10?

Answer. The average is 24/4 = 6. The sketch that returned 10 was off by 10 - 6 = 4.

Lecture 8 · 90 minutes

What you can now compute from a summary

You can now answer from a summary that fits in memory. In one pass you track a majority candidate on a stream of length 7, estimate counts in a sketch of 2 by 4 cells, and turn a trailing-zero record of 3 into a distinct-count estimate of 8. A reservoir of size 1 keeps each of 5 items with probability 1/5. Eight probes catch an array that is 1/4 ones, missing with probability about 0.100.

You can also place the same habit in parallel and in a second-moment sketch. Six tokens collapse to counts 3, 2, and 1 in one shuffle. Four random-sign sketches that return 4, 6, 10, and 4 average to the true second moment 6. The skill is to name the resource you spend, passes or probes or integers, and to attach the number the summary is allowed to get wrong, or to prove that on this instance it did not.

Name the resource, then the number you return.

1. Pick the summary

2. Spend probes or cells

3. Compute the estimate

4. State the allowed error

A sample probability beside a distinct estimate

1. A reservoir of size 1 over 5 items holds the third item with probability 1/5.

2. A trailing-zero maximum of 2 estimates 2 to the power 2 distinct items, which is 4.

3. One number is a sampling probability. The other is a distinct-count estimate. Both can be updated in the same pass.

Problem 1. Combine the reservoir with the trailing-zero rule. On 5 items the reservoir has size 1, and the sketch reports maximum trailing zeros 2. What is the probability it holds the third item, and what distinct-count estimate do you report?

Answer. The third item is stored with probability 1/5. The distinct-count estimate is 2 to the power 2, which is 4.

Problem 2. Combine count-min with the majority test. Count-min reports 4 for an item, and the stream has length 7. Can you conclude the item is a strict majority?

Answer. No. The report 4 is at least the true count, so the truth is at most 4 and might be smaller. A strict majority of 7 needs more than 3.5 occurrences, which an inflated 4 does not prove.